

Functional Annotation Report: *metE* (Q88JF1 / PP_2698) in *Pseudomonas putida* KT2440

Summary

The gene **metE** (locus tag **PP_2698**; UniProt **Q88JF1**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes **cobalamin-independent methionine synthase (MetE, EC 2.1.1.14)**, also called 5-methyltetrahydropteroyltriglutamate-homocysteine methyltransferase. This enzyme catalyzes the terminal, committed step of *de novo* methionine biosynthesis: the direct transfer of a methyl group from polyglutamylated **5-methyltetrahydrofolate (CH₃-H₄PteGlu₃)** to the thiol of **L-homocysteine**, producing **L-methionine** and **tetrahydrofolate (H₄folate)**. Unlike its cobalamin-dependent counterpart MetH, MetE performs this methyl transfer *without* a corrinoid (vitamin B₁₂) intermediate, relying instead on a catalytic **Zn²⁺** ion that activates the homocysteine thiol for direct nucleophilic (S_N2) attack on the N5-methyl group of the folate. The gene symbol, organism, catalytic annotation, Zn²⁺ cofactor, and conserved catalytic domain (Pfam PF01717 *Meth_synt_2*; InterPro IPR002629/IPR016456/IPR038071) are all mutually consistent — **the identity of this protein as a genuine MetE is well supported.**

A distinctive structural feature emerged from this investigation. Canonical bacterial and fungal MetE enzymes (e.g., *E. coli*, *T. maritima*, *S. mutans*, *C. albicans*) are large (~750–900 aa) **dual (β α)₈ TIM-barrel** proteins whose active site lies in the cleft between the two barrels, an architecture that arose by ancient gene duplication. Q88JF1, by contrast, is a compact **344-aa single-barrel ("archaeal-type") MetE**. A quantitative Smith-Waterman alignment performed during this investigation showed that ~90% of Q88JF1 aligns specifically and exclusively to the **C-terminal catalytic barrel** of *E. coli* MetE (residues 427–728; 28% identity, alignment score 303), with **no** alignment to the N-terminal folate-binding barrel. Crucially, the four resting-state zinc ligands (**His, Cys, Glu, Cys**) are fully conserved in Q88JF1 (e.g., Cys312 $\equiv$ *E. coli* Cys726), indicating that this streamlined enzyme retains a functional zinc/homocysteine active site.

Functionally, MetE operates in the **cytoplasm** within cysteine/methionine metabolism (KEGG pathway ppu00270, methionine-biosynthesis module M00017). It sits in a conserved, divergently transcribed **metE–metR** locus (the adjacent gene **PP_2696** encodes the LysR-type activator MetR), and *P. putida* KT2440 additionally encodes the B₁₂-dependent isozyme **MetH** (**PP_2375, Q88KB5**), giving the organism two routes to methionine depending on cobalamin availability. All functional conclusions for this specific *P. putida* protein are **inferential** — derived from sequence homology, conserved catalytic residues, genomic context, and structural studies of orthologs — because the UniProt protein-existence level is "4: Predicted" and no direct biochemical characterization of the *P. putida* enzyme has been published.

Key Findings

Q88JF1 is cobalamin-independent methionine synthase (MetE, EC 2.1.1.14)

The primary function of the gene product is unambiguous at the annotation level. UniProt Q88JF1 lists gene name *metE*, ordered locus name PP_2698, organism *P. putida* KT2440, and catalytic activity as **5-methyltetrahydropteroyltriglutamate–homocysteine methyltransferase**. The associated Gene Ontology terms are GO:0003871 (5-methyltetrahydropteroyltriglutamate-homocysteine S-methyltransferase activity), GO:0009086 (methionine biosynthetic process), and GO:0032259 (methylation). The listed cofactor is **Zn(2+)**.

The reaction catalyzed is the direct, single-step transfer of a methyl group from 5-methyltetrahydrofolate — in bacteria, the physiological substrate is the **poly-γ-glutamylated** form — to L-homocysteine, yielding L-methionine and tetrahydrofolate. This is the defining reaction of the MetE class: "*Cobalamin-independent methionine synthase (MetE) catalyzes the direct transfer of a methyl group from methyltetrahydrofolate to l-homocysteine to form methionine*" [PMID: 21840320]. The hallmark distinguishing MetE from cobalamin-dependent MetH — the absence of an intermediate methyl carrier — is captured by: "*Cobalamin-independent methionine synthase (MetE) catalyzes the transfer of a methyl group from methyltetrahydrofolate to L-homocysteine (Hcy) without using an intermediate methyl carrier*" [PMID: 15630480].

Substrate specificity: methyl donor = 5-methyltetrahydrofolate (physiologically polyglutamylated, CH₃-H₄PteGlu₃); methyl acceptor = L-homocysteine (its thiolate); products = L-methionine + tetrahydrofolate; obligate cofactor = a single catalytic Zn²⁺.

MetE is a Zn^{2+} -dependent methyltransferase with His-acid catalysis

The reaction chemistry is metal-assisted. The active-site Zn^{2+} functions as a **Lewis acid** that binds and deprotonates the L-homocysteine thiol, generating the reactive thiolate nucleophile. This nucleophile then performs an **in-line $\text{S}_{\text{N}}2$ attack** on the N5-methyl group of methyltetrahydrofolate. In the resting enzyme, the zinc is coordinated by the ligand set **His, Cys, Glu, Cys** — a signature distinct from the $(\text{Cys})_3$ zinc site of the cobalamin-dependent enzymes MetH and betaine-homocysteine methyltransferase (BHMT). A general-acid residue (His707 in *Thermotoga maritima* numbering) protonates the departing tetrahydrofolate leaving group through a water molecule trapped in the closed active site.

Supporting evidence: "*the MetE active site coordinates a zinc atom, which is thought to act as a Lewis acid and plays a role in the activation of thiol*" [PMID: 21840320]; the resting-state ligands are "*histidine, cysteine, glutamate, and cysteine in the resting form of MetE*" [PMID: 15630480]; and the closed-conformation structure "*reveals His707 as the acid that protonates the THF leaving group through a water molecule trapped in the closed active site. The substrates are correctly oriented for an in-line attack by l-homocysteine on the N(5)-methyl*" [PMID: 25545590].

Critically, Q88JF1 **retains these catalytic residues**: its sequence contains the **E-C-H** motif (...LECHNSH...) and the **C-x-x-C** motif (...PCTNCG...) that supply the zinc ligands of the C-terminal catalytic barrel, and UniProt lists $\text{Zn}(2+)$ as its cofactor with a "MetE C-terminal/archaeal" catalytic domain spanning residues 5–329.

Q88JF1 corresponds to the C-terminal catalytic barrel of a normally dual-barrel enzyme

Structurally characterized MetE enzymes are **dual $(\beta\alpha)_8$ TIM-barrel** proteins of ~750–900 residues. The N-terminal barrel arose by **gene duplication** of the C-terminal barrel and contributes folate-binding determinants, while the C-terminal barrel houses the catalytic zinc/homocysteine site; the active site sits at the interface between the two barrels. Q88JF1 is only **344 aa**, and its single annotated domain (residues 5–329) is the "Cobalamin-independent methionine synthase MetE C-terminal/archaeal" domain (Pfam PF01717; InterPro IPR002629). Q88JF1 is therefore a **single-domain, C-terminal-barrel (archaeal-type) MetE** rather than the canonical dual-barrel form.

Evidence for the canonical architecture and its evolutionary origin: "*The fold of the N-terminal barrel confirms that it has evolved from the C-terminal polypeptide by gene duplication*" [PMID: 15630480]; and "*The enzyme has a dual $(\beta\alpha)_8$ TIM barrel structure that binds, activates and*

brings the reactants into reaction proximity by conformational movements" [PMID: 25545590]. UniProt classifies the protein-existence evidence for Q88JF1 as "4: Predicted," meaning no direct experimental characterization of the *P. putida* protein has been reported.

Cytoplasmic enzyme in a conserved *metE*–*metR* locus, with a parallel B₁₂-dependent MetH route

The genomic context in *P. putida* KT2440 reproduces the classic **divergent *metE*–*metR* arrangement**. The immediate neighbor **PP_2696 (Q88JF3)** is the LysR/HTH-type transcriptional regulator **MetR**, which (modulated by homocysteine) transactivates *metE*. Separately, *P. putida* KT2440 encodes a full-length cobalamin-dependent methionine synthase **MetH (Q88KB5, PP_2375, 1235 aa)**, so the terminal methionine-synthase step can be performed by either MetE (B₁₂-independent) or MetH (B₁₂-dependent) depending on cobalamin availability. MetE is a **soluble cytoplasmic** protein with no signal peptide or transmembrane segment, consistent with a cytosolic biosynthetic enzyme.

Regulatory precedent from *E. coli*: "*The MetR protein is required for both metE and metH gene expression, and functions as a transactivator of transcription of these genes. MetR is a unique prokaryotic transcription activator in that it possesses a leucine zipper motif*" [PMID: 1943695]. B₁₂/MetH-mediated repression of *metE* is established: "*The repression of MetE synthesis in Escherichia coli by vitamin B12 is known to require the MetH holoenzyme*" [PMID: 1734876]. Under oxidative stress in *E. coli*, MetE is inactivated by **glutathionylation** of an active-site-entrance cysteine: "*Stoichiometric glutathionylation of MetE by GSSG occurs at cysteine 645, which is strategically located at the entrance to the active site*" [PMID: 15502870]. Functional interchangeability of the two synthases is illustrated by: "*Expression of the cobalamin-independent methionine synthase MetE alleviates this symbiotic defect*" in a system otherwise dependent on B₁₂ [PMID: 20698752].

Quantitative confirmation of single-barrel MetE identity and orthology assignment

A Smith-Waterman alignment (BLOSUM62; gap open –11, extend –1) of *P. putida* Q88JF1 (344 aa) against the *E. coli* canonical dual-barrel MetE (P25665, 753 aa) yielded a **score of 303 at 28.0% identity over a 314-residue aligned block**. Q88JF1 residues 6–314 (90% of its length) map onto *E. coli* MetE residues **427–728** — essentially the entire C-terminal catalytic (β α)₈ barrel — with **no significant alignment** to the *E. coli* N-terminal barrel (residues ~1–370). Because the *E. coli* catalytic zinc ligands (His641/Cys643/Cys726) fall within the aligned window, the zinc/homocysteine site is retained.

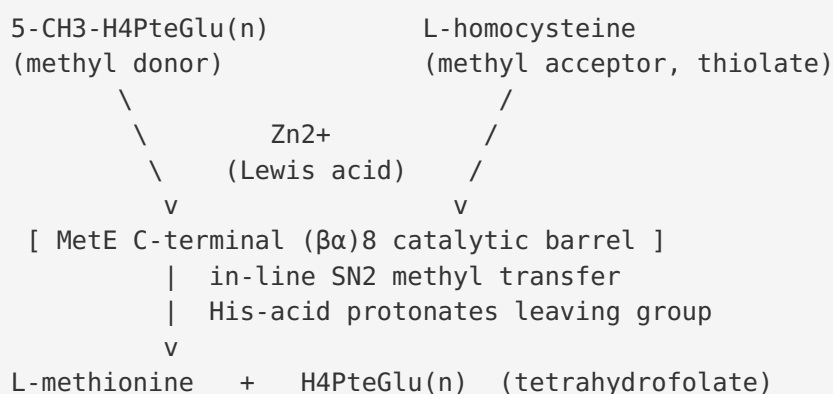
KEGG (ppu:PP_2698) independently corroborates the annotation: length 344 aa, genomic position complement(3089178..3090212) = 1035 bp, orthology **K00549** ("5-methyltetrahydropteroyltriglutamate--homocysteine methyltransferase [EC:2.1.1.14]"), pathway **ppu00270** (cysteine and methionine metabolism), module **M00017** (methionine biosynthesis, aspartate $\Rightarrow$ homoserine $\Rightarrow$ methionine). The flanking loci — **PP_2697** (flavin-reductase oxidoreductase) and **PP_2699** (DUF1852) — provide no candidate N-terminal-barrel gene, confirming that Q88JF1 is genuinely single-domain rather than a fragment of a split ORF.

Residue-level conservation of the zinc/homocysteine catalytic site

Traceback of the BLOSUM62 Smith-Waterman alignment maps the *E. coli* MetE C-terminal zinc-ligand **Cys726 directly onto Q88JF1 Cys312** (...LYPCTNCG...). Q88JF1 additionally contains the **...ISLECHNSH...** motif providing the Glu/Cys/His ligands corresponding to the *E. coli* His641/Glu642/Cys643 zinc-binding region (with the exact His/Cys registration shifting by a few residues owing to short loop indels at ~28% identity). Together, these place **all four resting-state MetE zinc ligands (His, Cys, Glu, Cys)** in Q88JF1 at the correct topological positions of the C-terminal catalytic barrel — the strongest single line of evidence that this compact protein is a *functional*, not merely homologous, methionine synthase.

Mechanistic Model / Interpretation

The reaction



Step-by-step chemistry: 1. The catalytic Zn^{2+} binds L-homocysteine and lowers the pK_a of its thiol, generating a reactive **thiolate** nucleophile. 2. The thiolate performs an **in-line $\text{S}_\text{N}2$ attack** on the electrophilic **N5-methyl carbon** of 5-methyltetrahydrofolate. 3. A conserved **His general acid** (His707 in *T. maritima*) protonates the N5 nitrogen of the departing tetrahydrofolate through a trapped active-site water, facilitating C–N bond cleavage. 4. Products **L-methionine** and **tetrahydrofolate** are released; conformational "closing" of the barrel during catalysis excludes bulk solvent and aligns the reactants.

Structural distinction: single vs. dual barrel

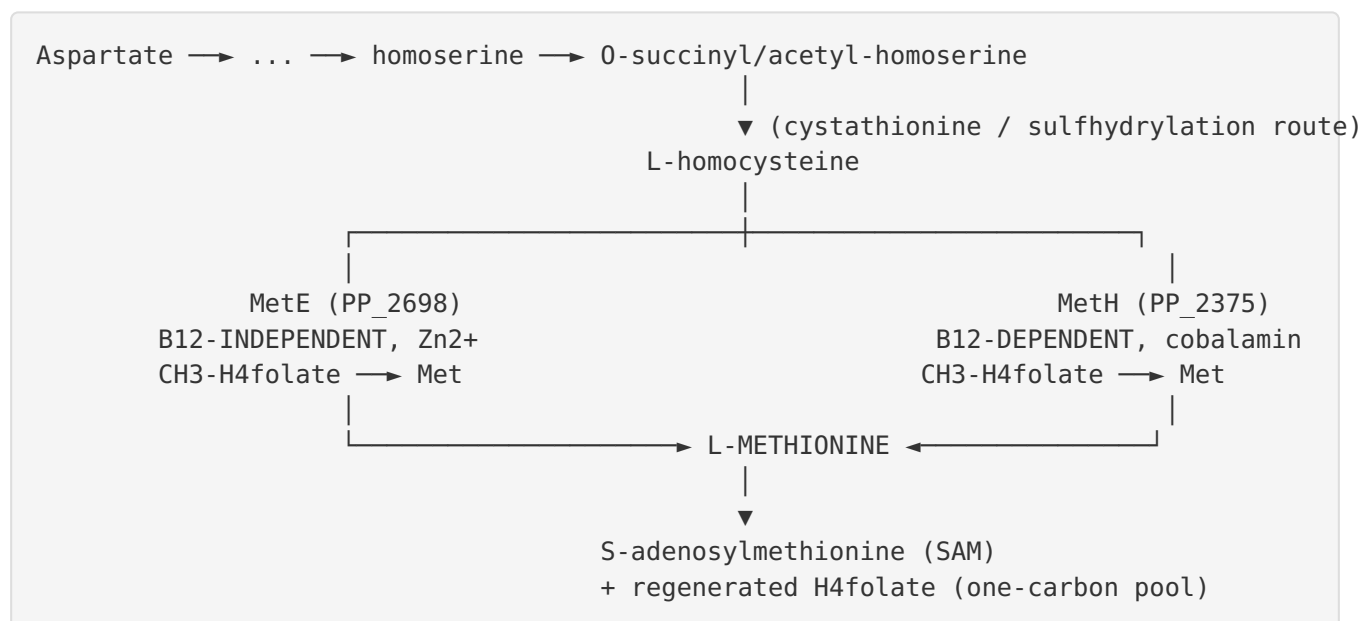
Property	Canonical MetE (<i>E. coli</i> , <i>T. maritima</i> , <i>C. albicans</i>)	Q88JF1 (<i>P. putida</i> PP_2698)
Length	~750–900 aa	344 aa
Architecture	Dual $(\beta\alpha)_8$ TIM barrel	Single $(\beta\alpha)_8$ barrel (C-terminal/archaeal type)
N-terminal barrel	Present; folate-binding, from gene duplication	Absent
C-terminal catalytic barrel	Present; houses Zn^{2+} /Hcy site	Present and conserved
Zinc ligands (resting)	His, Cys, Glu, Cys	His, Cys, Glu, Cys (all retained)
Zinc mechanism	Lewis-acid thiol activation	Inferred identical
Active-site location	Between the two barrels	Single barrel ($\pm$ oligomerization)

The absence of the N-terminal barrel is the report's most notable structural finding. In dual-barrel enzymes, the N-terminal barrel contributes folate-binding residues (e.g., Asn126 in *C. albicans* Met6p, whose mutation abolishes activity —).

P 24524835

In a single-barrel archaeal-type MetE, folate-binding determinants must be supplied by the C-terminal barrel itself (and possibly by oligomerization), consistent with the existence of functional single-barrel MetE enzymes in archaea and some bacteria.

Pathway and physiological context



MetE regenerates tetrahydrofolate, feeding the cellular **one-carbon (folate) pool**, and supplies methionine for protein synthesis and for **S-adenosylmethionine (SAM)** — the universal methyl donor. Its transcription is activated by **MetR (PP_2696)** in response to homocysteine, and its physiological importance is greatest under **B₁₂-limiting conditions**, where MetH cannot function. In *E. coli*, MetE is also a redox sensor: **glutathionylation of an active-site cysteine** during oxidative stress inactivates it, transiently converting the cell to methionine auxotrophy — a plausible regulatory feature for the *P. putida* enzyme as well.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
21840320	<i>S. mutans</i> MetE crystal structures; zinc-inversion model	Defines the MetE reaction; establishes Zn ²⁺ as the Lewis-acid thiol activator
15630480	<i>MetE: a face-to-face double barrel that evolved by gene duplication</i>	Confirms direct methyl transfer without a carrier; defines His/Cys/Glu/Cys zinc ligands; documents dual-barrel gene-duplication origin
25545590	<i>MetE captured in a substrate-induced closed conformation</i>	Describes His-acid catalysis, in-line S _N 2 attack, and the dual (β α) ₈ barrel conformational mechanism
24524835	<i>Structural analysis of a fungal methionine synthase (Met6p)</i>	Two-barrel architecture; catalytic zinc positioning on homocysteine binding; folate-binding residues (e.g., Asn126) essential for activity
15502870	<i>Oxidative stress inactivates MetE in E. coli</i>	Redox regulation: glutathionylation of active-site-entrance Cys645 inactivates MetE
1943695	<i>Regulation of methionine synthesis in E. coli</i>	MetR is the transcriptional activator of <i>metE</i> — supports the divergent <i>metE</i> – <i>metR</i> locus (PP_2696)
1734876	<i>metF/metJ genes and B12 regulation of methionine genes</i>	B ₁₂ /MetH-mediated repression of <i>metE</i> — the MetE↔MetH regulatory relationship
20698752	<i>S. meliloti cobalamin-dependent RNR for symbiosis</i>	MetE can functionally substitute for B ₁₂ -dependent methionine synthesis

All eight papers are structural/biochemical or regulatory studies of MetE orthologs and the methionine-biosynthesis regulon. **None studies the *P. putida* KT2440 protein directly**, so the evidence base is homology- and orthology-driven rather than target-specific. Database annotations (UniProt Q88JF1, KEGG ppu:PP_2698, Pfam PF01717, InterPro IPR002629/IPR016456/IPR038071) provide the target-specific anchor, corroborated by the quantitative alignment and residue-conservation analyses performed in this investigation.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of Q88JF1.** UniProt protein-existence level is "4: Predicted." No published enzymatic assay, crystal structure, or knockout phenotype exists for the *P. putida* KT2440 protein specifically. All functional conclusions are inferential.
2. **Modest sequence identity (~28%) to characterized enzymes.** Although the zinc-ligand set and catalytic barrel topology are conserved, the exact His/Cys registration shifts by a few residues due to short loop indels; direct structural confirmation would remove residual ambiguity.
3. **Folate binding in a single-barrel context is untested.** In dual-barrel MetE, the N-terminal barrel supplies folate-binding residues. How the single-barrel *P. putida* enzyme binds polyglutamylated methyl-THF — via C-terminal-barrel residues, an oligomeric interface, or a different geometry — is unknown.
4. **Oligomeric state unknown.** Single-barrel archaeal-type MetE enzymes may dimerize to reconstitute a composite active site; this has not been examined for Q88JF1.
5. **Regulatory details inferred from *E. coli*.** MetR-mediated activation and B₁₂/MetH repression are established in *E. coli*; the precise *P. putida* KT2440 regulatory logic (operator sequences, homocysteine co-activation, B₁₂ responsiveness) has not been demonstrated experimentally.
6. **Redox regulation not verified in *P. putida*.** Whether the *P. putida* enzyme is glutathionylation-inactivated like *E. coli* MetE depends on the presence/position of an active-site-entrance cysteine that was not explicitly mapped here.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and activity assay.** Clone PP_2698, express in *E. coli*, purify, and measure methyltetrahydrofolate–homocysteine methyltransferase activity (e.g., the fluorescence-based assay from

P 24524835

confirming Zn²⁺ dependence and B₁₂ independence. Determine K_m for CH₃-H₄PteGlu(n) and L-homocysteine, and test polyglutamate specificity.

2. **Zinc-content and metal-substitution analysis.** Quantify bound Zn^{2+} (ICP-MS) and test $\text{Co}^{2+}/\text{Cd}^{2+}$ substitution to confirm the Lewis-acid role, alongside site-directed mutagenesis of the predicted ligands (His, Cys312, the ECH-motif residues).
3. **Structural determination.** Solve the crystal or cryo-EM structure of Q88JF1 to confirm the single-barrel fold, define how folate is bound without an N-terminal barrel, and establish the oligomeric state. An AlphaFold model can provide an immediate low-cost prediction of the fold and active-site geometry.
4. **Genetic complementation.** Test whether PP_2698 complements an *E. coli* ΔmetE methionine auxotroph, and construct a *P. putida* ΔPP_2698 mutant to assess methionine auxotrophy under B_{12} -free vs. B_{12} -supplemented conditions (probing MetE/MetH redundancy).
5. **Regulatory validation.** Use RNA-seq/qPCR and a MetR (PP_2696) knockout to confirm homocysteine-dependent, MetR-activated transcription and B_{12} -mediated repression of PP_2698 in *P. putida*.
6. **Oxidative-stress sensitivity.** Test whether purified Q88JF1 is inactivated by GSSG (glutathionylation) and map any reactive active-site-entrance cysteine, to determine whether the *E. coli*-style redox switch is conserved.

Report generated from a 3-iteration autonomous investigation. Findings are supported by domain/annotation databases (UniProt, KEGG, Pfam/InterPro), quantitative sequence alignment against characterized orthologs, and the structural/regulatory literature cited above. The identity of PP_2698 as a genuine, single-barrel, cobalamin-independent methionine synthase is well supported; its detailed biochemistry remains to be confirmed experimentally.
